Benchmark design governs reported skill in daily river water-temperature prediction

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Key points

- A seven-day skill of +0.250 against persistence falls to +0.038 against a damped-persistence reference at 116 reportable gauged sites.
- A gradient-boosted tree with site identity is most accurate at 1- and 3-day leads; the deep model matches it at 7 days.
- Learned skill beyond the seasonal reference concentrates in rapid warming and rapid cooling states.

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Abstract

Daily river water temperature is strongly persistent and strongly seasonal, and machine-learning studies routinely report large skill gains for it, usually measured against naive persistence or climatology on random data splits with each model scored on its own set of predictable days. Such designs cannot separate learned river behavior from cheaper explanations: seasonal damping, favorable key selection, and spatial interpolation between neighboring gauges. Here we evaluate a constrained deep predictor against damped persistence, gradient-boosted trees with and without site identity, a global LSTM, an information-matched plain causal convolutional network, and an empirical thermal-recurrence comparator on one common registry of station/date/horizon keys at 120 U.S. gauges (116 reportable on the 2021–2023 test window) spanning 15 hydrologic regions, with all preprocessing fitted strictly backwards in time; the primary evaluation is a known-site temporal holdout, and a separate matched experiment holds out whole regions rather than random sites. On the independent 2021–2023 window the deep model reports a seven-day station-median skill of +0.250 against persistence but +0.038 against damped persistence; the median station-level memory gain is 0.49 °C and the median learned gain 0.07 °C at seven days, and the median fraction of the station-level error reduction delivered by seasonal memory is 0.875. A gradient-boosted tree with site identity has the lowest station-median RMSE at one and three days (0.589 and 1.304 °C) and is matched at seven days (1.735 versus 1.694 °C; the paired difference is not distinguishable from zero at the whole-HUC2 cluster level, Table 4.6). When the held station's own thermal history remains available, whole-region transfer incurs a small, consistently signed additional error of about 0.01 °C relative to random-site transfer — an order of magnitude smaller than the reference-model effect — and the learned model's residual value concentrates in rapid warming and rapid cooling states (-0.30 and -0.18 °C station-median paired ΔRMSE at seven days); withholding the station's thermal history from preprocessing instead costs about 0.2 °C at seven days in the matched comparison, so the spatial partition is secondary specifically given local thermal memory. Reported skill for this variable is therefore largely a statement about the reference model and the evaluation design rather than about the architecture.

Plain Language Summary

River temperature changes slowly from day to day, so a forecast that repeats yesterday's reading is already fairly accurate, and one that also nudges it toward the usual value for the time of year is better still. We tested a river-temperature model at 120 U.S. gauges (116 reportable on the held-out test window) and deliberately made the evaluation demanding: every model was scored on exactly the same days and sites, none could see information from after the moment it was asked to predict, and in a separate experiment we held out entire river regions rather than scattered gauges. Measured against a simple seasonal adjustment of yesterday's reading, most of the advantage that a naive comparison would report disappeared, and a standard tree-based method was more accurate at the shortest forecast ranges. How well such a model performs depends mostly on what it is compared against and, to a smaller and measurable degree, on how the map is divided rather than on details of the model itself.

Keywords: river water temperature; benchmark design; strong baselines; spatial holdout; comparative evaluation; conformal prediction.

1 Introduction

Stream and river thermal regimes set dissolved-oxygen saturation, metabolic rates, life-stage timing, and habitat suitability, and they respond to atmospheric forcing, discharge, groundwater exchange, riparian shading, channel geometry, and regulation (Caissie, 2006). Water managers need daily-resolution temperature at gauged reaches to time releases, anticipate thermal refuge loss, and interpret compliance monitoring, and the supply of daily records from national monitoring networks has made the variable one of the most heavily modeled targets in applied hydrological machine learning.

The prevailing understanding in that literature is that deep sequence models have substantially advanced daily river-temperature prediction. Nonlinear air–water regressions (Mohseni et al., 1998) and hybrid air-temperature and discharge formulations (Toffolon and Piccolroaz, 2015; Piccolroaz et al., 2016) were succeeded by recurrent, multi-task, and physics-guided river-network architectures that report large error reductions (Feigl et al., 2021; Rahmani, Lawson, et al., 2021; Jia et al., 2021; Sadler et al., 2022; Zwart et al., 2023; Barclay et al., 2023). The reported gains are consistent enough that architectural sophistication is a common explanation for accuracy in this problem. Graph-convolutional and temporal-graph architectures propagate information along the river network and so make spatial dependencies explicit, and the relationships they learn have been interrogated with explainability methods for how they govern transfer to unseen environmental conditions (Sun et al., 2021; Topp et al., 2023). That transfer is a question of spatial sensitivity rather than of raw accuracy, and it is the one this study isolates by holding out whole regions rather than neighbouring sites.

Those reported gains are, however, extraordinarily heterogeneous, and the heterogeneity does not organize itself by architecture. Reviews of the field note that published accuracies are not comparable across studies because reference models, key sets, and data-splitting conventions differ (Corona and Hogue, 2025). The dispersion is larger than the architectural differences it is supposed to explain, and it survives within single panels: on the data used here, one fixed model and one fixed set of prediction days produce a seven-day skill of +0.251 or of +0.038 depending only on which reference model is placed in the denominator. An architectural account cannot produce that spread, because the architecture does not change between the two numbers.

The identification gap is a property of the designs, not of the literature's size. Three design choices recur in this literature, and each one absorbs an alternative explanation into the reported number. First, skill is quoted against naive persistence, climatology, or an air–water regression. Daily mean water temperature has large thermal inertia, so relaxing the last observation toward a seasonal climatology already removes most of the error a learned model can remove; a reference that omits this hands the model a large part of its budget before it learns anything. Second, models are scored on model-specific complete-case sets, so a model can gain apparent accuracy by declining to predict on hard days, and preprocessing statistics — scaling constants, imputation fills, climatological references, event thresholds, interval offsets — are commonly fitted on periods that include the evaluated interval, which transfers information backwards in time without leaving a trace in a conventional split. Third, spatial generalization is reported from random held-site splits. When a held-out gauge's neighbours remain in training, the model reaches the held-out regime through correlated forcing and spatially smooth representations; the quantity measured is closer to interpolation than to transfer. None of these choices is detectable from a reported RMSE, and each can inflate it.

Here we hold the model and the data fixed and vary the three design choices instead, so that their separate contributions to reported skill become measurable. We assembled a panel of 657,480 site-days from 120 stable U.S. Geological Survey site numbers spanning 2006–2020, 34 states, and 15 two-digit hydrologic unit (HUC2) regions, tuned all models on data through 2020, and evaluate them on one common set of station/date/horizon keys at 1-, 3-, and 7-day leads, with a held-out 2021–2023 test window. Four questions organize the study. How much of a reported gain survives a strong reference? Which model class is actually most accurate on identical keys? How much of a reported spatial transfer survives whole-region holdout, once local adaptation is separated from spatial geometry? And in which issue-time-identifiable hydrologic states does a learned model still add skill beyond the strong reference? The predictor whose skill we decompose, ThermoRoute, is described in full in Section 3.1; its architecture is the object under test rather than the contribution.

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2 Data and design

2.1 The frozen panel and the cohort it defines

The development record is a single derived panel, `data_usgs/panel_usgs_120v2.parquet`, bound by a manifest (`data_usgs/frozen_panel_v1.json`) to a station registry (`data_usgs/station_registry_v1.csv`). Each of 120 sites carries one row for every calendar day from 2006-01-01 through 2020-12-31, giving 657,480 rows with no gaps in the date index; missing observations are explicit nulls rather than absent rows. Station identity is the zero-padded USGS `site_no`.

The seven raw variables consumed by the models, in frozen order, are water temperature (`WTEMP`), streamflow (`FLOW`), air temperature (`TEMP`), precipitation (`PRCP`), a relative-humidity proxy (`RHMEAN`), Daymet daylight-period mean incoming shortwave radiation in W m^{-2} (`DH`), and wind speed (`WDSP`). Two names are legacy and easily misread: `DH` is Daymet `srad`, the incident shortwave flux averaged over the daylight period, not day length and not a 24-hour mean; `RHMEAN` is a reproducible vapour-pressure proxy evaluated at the tmax/tmin midpoint, not a direct daily-mean humidity observation. Gage height (`WLEVEL`) is retained as raw evidence but is not a model input. Water temperature and discharge come from USGS NWIS daily values (U.S. Geological Survey, 1994), meteorological fields from Daymet V4 (Thornton et al., 2022), and wind from gridMET (Abatzoglou, 2013).

The cohort was not sampled at random from U.S. rivers, and nothing here treats it as though it were. Candidate selection began from a discovery query over stream sites in the states represented by the panel and rejected 1,345 of 1,465 candidate stations: 950 lacked joint water-temperature and discharge availability, 376 failed a full-period coverage threshold, and 19 failed a 2019–2020 coverage threshold. The retained cohort is therefore availability-enriched, and the 2019–2020 interval participated in its construction, so that interval is development data in the strict sense and is used for model tuning and diagnosis rather than as an independent test.

Three properties of the retained cohort bear directly on the analysis. Observed water temperature is absent on about 15.8% of panel rows, with a maximum contiguous gap of 2,404 days, and discharge on about 2.8%, with a maximum gap of 1,369 days; the meteorological fields are missing on about 0.07% of rows, in a structured rather than random way, because all 120 sites lack the provider calendar's final day in the leap years 2008, 2012, 2016, and 2020. Among retained stations, 38 share a repeated hydrologic unit code with another retained station and 19 have another retained station within 10 km, so station-level independence is not tenable. Two stations contain 2,059 rows of signed negative discharge (minimum -121 cfs), retained with their sign rather than clipped.

2.2 Temporal roles and the common key set

Temporal roles are fixed before any model is fitted, and a training sample is admitted only if its issue date and every one of its target dates fall inside the same partition.

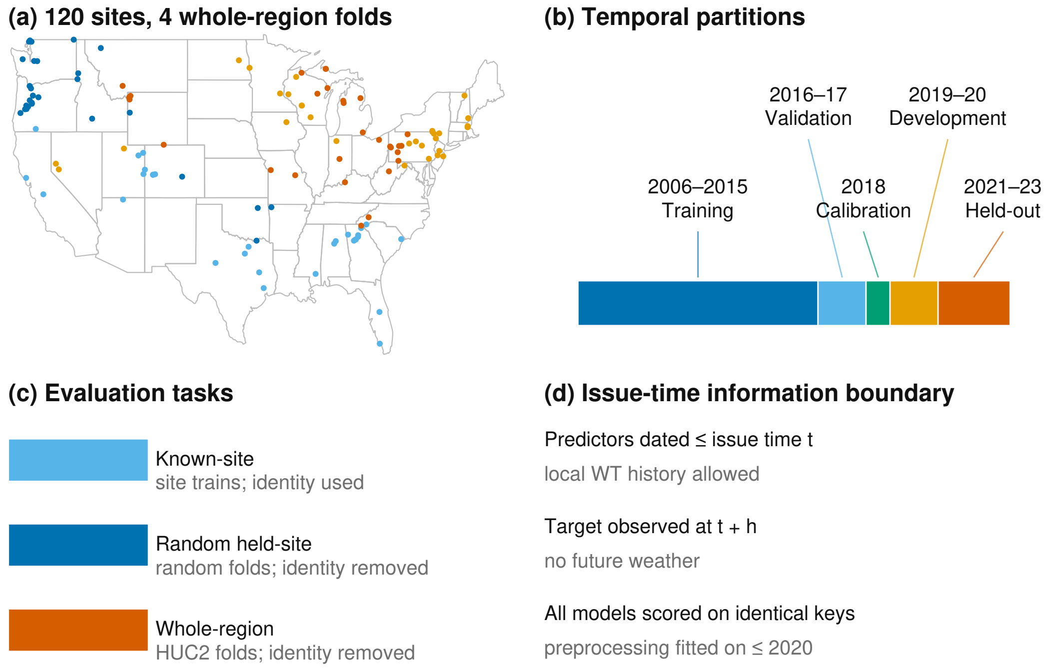

Figure 1. Study sites and evaluation design.

Role	Dates	Rows	Observed `WTEMP`	Permitted use
Training	2006–2015	438,240	341,646	fit preprocessing and models
Validation	2016–2017	87,720	84,074	select model settings
Calibration	2018	43,800	42,279	conformal and Platt fits; final year of the 2006–2018 seasonal event reference
Development evaluation	2019–2020	87,720	85,621	model tuning and diagnosis
Held-out test	2021–2023	—	observed	comparative evaluation (Section 4.4)

Models are tuned exclusively on data through 2020; the 2021–2023 window is held out and used only for the comparative evaluation of Section 4.4. On the development-evaluation partition, the intersection of admissible keys across all primary models comprises 249,072 station/date/horizon keys, evenly distributed as 83,024 keys per lead; the held-out window carries its own common-key registry of the same form (Section 4.4). The 249,072 figure describes the development registry only and is never used for a held-out claim.

Figure 1. Study sites and evaluation design. (a) The 120 U.S. gauges colored by the four deterministic whole-HUC2-region folds used for the spatial-transfer experiment, on a CONUS outline; each fold holds out whole regions rather than scattered sites. (b) Temporal partitions: 2006–2015 training, 2016–2017 validation, 2018 calibration, 2019–2020 development evaluation, and 2021–2023 independent held-out test. (c) The three evaluation tasks: known-site forecasting (site identity used), random held-site transfer, and whole-region gauged transfer (site identity removed; local water- temperature history remains an allowed issue-time input in every task). (d) Issue-time information boundary: predictors are dated no later than the issue date, targets are observed at t + h, and all models are scored on the identical common forecast-key registry, with preprocessing fitted on data through 2020 only. Every development comparison in Sections 4.1–4.3 uses exactly this key set. No model-specific complete-case set is permitted, so a model cannot gain apparent accuracy by declining to predict on hard keys. The held-out 2021–2023 evaluation uses the same admissibility rule applied to the later window.

2.3 Evaluation-period inputs and the information boundary

The test interval is 2021-01-01 through 2023-12-31 for the same 120-site cohort, and historical Daymet and gridMET covariates for that interval are retrieved and archived. Meteorology is represented at each station coordinate and is not aggregated over upstream catchments, which is a real limitation for large basins (Section 6.2).

The primary information set uses provider values dated no later than each historical issue date and consumes no horizon-specific future weather forecast. This is a date-indexed retrospective hindcast, and not a re-execution of what a forecaster could have run on the day (Section 6.1): the as-issued provisional vintage of a gridded product cannot be reconstructed after the fact, so archiving requests, responses, timestamps, and checksums freezes the dataset actually evaluated without proving that identical values were available operationally at the time. NWIS dates denote site-local finalized daily values while Daymet and gridMET use provider-specific calendar-day definitions; no subdaily day-boundary harmonization is claimed. A product-compatibility bridge re-fetched 2018–2020 covariates with the parser used for the test-window acquisition and records `PASS_EXACT_PRODUCT_BRIDGE` on the exact site/date registry.

Daily mean water temperature (00010, °C), discharge (00060, cfs), and raw-only gage height (00065, ft) for 2021–2023 are retrieved by direct requests to the USGS NWIS waterservices daily-values REST service (`https://waterservices.usgs.gov/nwis/dv/`), with every request and response byte-pair recorded, checksummed, and content-addressed by a `SnapshotStore` cache, preserving qualifiers, timestamps, and content hashes. Where two or more finite series exist for a parameter and date, the cell is marked missing with an explicit conflict code rather than averaged or selected. The primary analysis uses every finite parsed value regardless of approval qualifier, so qualifiers can never select or remove a station, model, date, or forecast key; a separately labeled sensitivity retains target keys only when the target qualifier has the exact token set {`A`}. Test-window daily means are outcomes only; they are not read by model-selection, feature-selection, threshold-selection, calibration, or station-inclusion code, all of which is fixed on data through 2020.

A pooled-preprocessing sensitivity re-fits the model suite on the same 120-site registry with station-agnostic (pooled) transforms in place of the per-station transforms used by the primary arm, so that preprocessing aggregation is the only difference between the two. It is not a site-disjoint cohort: every site in the sensitivity arm is already in the development registry, and each target site still supplies its own observed history (Section 3.4).

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3 Methods

3.1 The predictor under test

ThermoRoute predicts at issue time t, station i, and lead h by adding a bounded learned residual to a damped-persistence anchor,

$$\textbf{(1)}\ A_{i,t+h} = c_{i,t+h} + \phi_i^h\,(y_{i,t} - c_{i,t})$$

where c is a seasonal climatology and ϕ_i a per-station decay fitted by no-intercept least squares on training-period consecutive-day anomaly pairs (at least 30 pairs, clipped to [0, 0.999]). The same object, fitted identically, is the damped-persistence reference.

The residual is produced by a strictly left-looking temporal convolutional encoder over a 14-lag window, with a horizon-conditioned variable/lag router, a three-expert mixture, and a $\pm 1\,^{\circ}$C algebraic bound on the deviation from the anchor. One model serves all three leads; 38,505 trainable parameters; five seeds averaged. The full specification, loss, and fitting protocol are in Text S3. The architecture is the object under test, not the contribution: Section 4.3 shows an information-matched plain causal network reproduces it to within 0.023 °C, and Section 4.4 quantifies what the information set, not the architecture, is worth.

3.2 The reference set

The composition of the reference set is the most consequential methodological choice in this study, because it is what the reported gain is a gain over. Each member removes one alternative explanation for apparent skill.

Persistence (ŷ_{t+h} = y_t) quantifies the raw memory of the series. **Damped persistence**, equation (1) fitted on the training period, absorbs the two effects a learned model most easily rediscovers — inertia and seasonality — and is the reference against which the primary comparisons are stated. **Seasonal climatology** isolates the seasonal cycle alone. **LightGBM** (Ke et al., 2017) is the principal learned reference and is run both as a global model receiving stable site identity as a categorical feature and as a per-station variant that isolates the value of pooling. A **global LSTM** with a station embedding represents the deep sequence family that has produced the strongest recent results for this variable (Rahmani, Lawson, et al., 2021; Zwart et al., 2023). A **plain causal temporal convolutional network** is the information-matched deep reference: it receives the same outcome-free issue-time history, station identity, climatology, frozen damped anchor, standardized environmental context, and calendar and regime-gate tensors as the full model, is matched to within 2% of its parameter count (38,346 against 38,505) and to the same optimiser-step budget, and predicts an unrestricted residual around the same anchor while omitting the relaxation proposal, router, mixture, and residual bound. Its companion, a plain multilayer perceptron at 38,860 parameters, isolates the value of sequence structure. Neither ever reads the target tensor, the target-date field, or the disabled water-level channel.

The learned references are given a genuine chance to win. The global LSTM uses the same development splits, loss, station-balanced sampling, epoch and patience limits, history length, site identity, and five-seed budget as ThermoRoute, its validation-only tuning grid may expose the same climatology, damped-anchor, and season features, and it is eligible for the same calibration; it lacks only the bounded-residual constraint and the lag router. LightGBM selects among four candidate settings separately by lead on the 2016–2017 partition using station-macro RMSE alone; the LSTM selects among three candidate architectures with seed 0 before fitting five members. Tuning budgets are documented but *not* equalized across model classes, which is a real asymmetry: any statement about LightGBM here is scoped to this four-candidate procedure and this feature schema, not to gradient boosting in general. Seven one-factor architecture controls (`DampedPriorOnly`, `TR-noDynamicPrior`, `TR-fixedKappa`, `TR-noRouter`, `TR-noMoE`, `TR-noTCN`, `TR-unbounded`) are fitted with the same five seeds and paired within seed on identical keys before ensemble averaging; they are deletion and intervention sensitivities and do not prove component necessity or identify a mechanism. `DampedPriorOnly` disables both the dynamic prior and the residual model, so its output is identically the damped-persistence anchor; it reproduces the `DampedPersistence` baseline to within floating-point noise on every metric and is therefore not reported as a separate row in the held-out tables.

An air2stream-style hybrid reference (Toffolon and Piccolroaz, 2015) is fitted per station on the training record and scored on the held-out window with the same keys as every other model (Section 4.4; Supporting Information SI07). It is an unofficial empirical thermal-recurrence variant of the published formulation — not the official code and not a validated reproduction — so no process-side claim is attached to its scores (Section 6.2).

3.3 Leakage control

Leakage control here is a set of mechanisms that can be checked against the code and the artifacts, rather than a statement of intent.

Issue-time separation of covariates. Every predictor consumed by a primary model carries a date no later than the issue date, and no horizon-specific future weather field enters any primary model.

Non-anticipating sequence encoding. Equation (5) makes the information horizon bounded and auditable by construction rather than by convention.

Strictly backward-fitted statistics. Standardization constants, imputation fills, the seasonal climatology, the damped-persistence rate, the station-level q90 event thresholds (2006–2015), the seasonal event reference (2006–2018), the conformal offsets (2018), and the Platt calibrators (2018) are each fitted on data strictly preceding the interval on which they are applied. This closes the most common backward-information channel in hydrological benchmarking, which is not the model but the preprocessing.

Explicit admissibility. A forecast key is admissible only when issue-date water temperature is genuinely observed and a 32-day history can be constructed; other history cells are filled with training-only seasonal medians and retain explicit missingness masks. We record one honest gap: the issue-date auxiliary quantities used by the relaxation proposal and the regime gate are computed from the same training-only imputed panel but do not each carry a separate auxiliary-path mask, so the output is not guaranteed invariant to the chosen fill values.

Reproducibility check. As a reproducibility check, a fresh interpreter reloads every trained member and reproduces the validation, calibration, and 2019–2020 keys and values from the frozen inputs, independently recomputing the thresholds and refitting the calibration parameters; a prediction file that is merely self-consistent is rejected. The details of that replay are specified in the Supporting Information.

3.4 Two spatial partitions, separated from local adaptation

Random held-site splits are the usual way to report spatial generalization in this literature, but they can produce optimistic transfer estimates: if a held-out gauge's neighbours remain in training, the model reaches the held-out site's thermal regime through correlated forcing, shared preprocessing statistics, and spatially smooth learned representations. We therefore report two arms with different spatial information properties and read the difference between them as the measurement of interest, in a 2×2 factorial that separates the spatial partition from the local-adaptation policy (protocol v1; Section 4.7 and SI11).

The **random held-site warm-start** arm uses four balanced folds in which held-site histories still contribute to global panel preprocessing. This is the arm most comparable to published random-split results, and precisely for that reason it is not the arm we treat as the spatial test.

The **held-region gauged transfer** arm packs the 15 HUC2 groups into four folds of [30, 30, 31, 29] stations and holds out *whole regions*, so no gauge from a held-out region appears in training. Station-agnostic ThermoRoute and a global LightGBM are each trained on the in-fold regions and forecast the held-out region's stations, with all climatology, scaling, and damped-rate parameters pooled from in-fold stations only. The mean distance from a held-out station to its nearest training gauge is 289 km.

The factorial contrasts two adaptation policies within each geometry. *Local* adaptation lets each target station's own 2006–2015 history contribute to its climatology, damped anchor, and imputation medians — the definition of gauged transfer used throughout this paper. *Pooled* adaptation fits those statistics over the in-fold training stations only, so a held region's long histories never enter preprocessing; the contrast between the two isolates the value of local thermal history separately from the spatial partition. On the independent 2021–2023 window the experiment is run with a station-agnostic gradient-boosted tree (raw and damped-residual targets) over four region folds and five repeated random-split seeds (Section 4.5); the development-period ThermoRoute arm of the spatial analysis is archived in SI19.

The label on this arm must be exact. Held-out sites still supply their own observed water temperature through the issue date, both as the persistence anchor and as the sequence history. This is **gauged** transfer to a region whose gauges were not used in fitting; it is **not** prediction at a site with no observational record, which is a different problem and a different evaluation (Weierbach et al., 2022; Rahmani, Shen, et al., 2021).

3.5 Conformal intervals

Split-conformalized quantile intervals are calibrated on the 2018 year only and are reported as an empirical marginal coverage diagnostic with width and interval score beside every coverage figure; no finite-sample or conditional-coverage claim is made (details and sensitivities in SI08).

3.6 Estimand, metrics, and comparison set

Two differently signed quantities are reported, and they are never combined. The paired effect is

(2) ΔRMSE = RMSE(candidate) − RMSE(reference), in °C, **negative** favors the candidate

and the skill score is

(3) skill = 1 − RMSE(candidate)/RMSE(reference), dimensionless, **positive** favors the candidate

Equation (9) is the formal estimand and carries a °C unit everywhere it appears; equation (3) is a ratio and appears as a bare signed number. Every value in Section 4 is labeled with which of the two it is. A positive ΔRMSE and a positive skill score mean opposite things, and no figure axis, table column, or sentence in this paper mixes them.

The sampling unit is the station. For each lead, unweighted RMSE is computed on the common daily keys separately for each reportable station, and the primary effect is the unweighted median across stations of equation (2) with ThermoRoute as candidate; a station/lead cell is reportable only with at least 100 valid paired targets, so daily rows do not determine between-station weight. On the held-out 2021–2023 window we additionally report mean absolute error (MAE) and mean error (bias), and skill against both persistence and seasonal climatology, per model and lead, both pooled and per station. We report RMSE and paired RMSE differences rather than an efficiency-type criterion such as the Nash–Sutcliffe efficiency or its decomposition-based successors (Nash and Sutcliffe, 1970; Gupta et al., 2009), because those criteria normalize by a station-specific variance, which would make a paired between-model difference on identical keys harder to read.

The comparison set contains five head-to-head rows:

#	Comparison	Lead	Reference margin
1	ThermoRoute vs. damped persistence	1 d	0.00 °C
2	ThermoRoute vs. damped persistence	3 d	0.00 °C
3	ThermoRoute vs. damped persistence	7 d	0.00 °C
4	ThermoRoute vs. LightGBM	3 d	+0.05 °C numerical ceiling
5	ThermoRoute vs. LightGBM	7 d	+0.05 °C numerical ceiling

The +0.05 °C ceiling is a stated numerical reference for the LightGBM comparison rather than a decision threshold: it is not derived from sensor precision, biological response, a water-quality standard, or an elicited stakeholder utility, and it carries no ecological or regulatory importance.

Two uncertainty procedures accompany each row, both clustered at HUC2: a one-sided p-value from exact enumeration of all 2^K whole-cluster sign vectors, applying one common sign to every station effect within a cluster, and a 10,000-draw whole-HUC2 cluster bootstrap percentile interval for the median station effect, with Holm adjustment (Holm, 1979) over these five p-values. We do not use the standard tests of equal predictive accuracy for forecast series (Diebold and Mariano, 1995; Harvey et al., 1997), because their asymptotics are stated for a single loss-differential series and do not address dependence across the 120 gauges, which is the dominant dependence here.

These clustered procedures rest on assumptions the cohort only partially meets, and the limit is stated rather than hidden. The cohort was assembled by an availability filter, not by probability sampling of HUC2 regions, so exchangeable region sampling is not established; with 15 HUC2 groups (an inverse-Herfindahl effective count of about 9.54 and a largest group holding 21.7% of stations) the clustered intervals and p-values are approximate descriptive sensitivities, not decision evidence. No comparison here is written as a superiority, non-inferiority, equivalence, or parity statement, and none generalizes to a national population. Equal-HUC aggregation, leave-one-HUC2 influence, and year-by-season temporal-stability audits are reported as held-out sensitivities; HUC2 is a coarse administrative grouping, not an independent river-network component.

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4 Results

Sections 4.1–4.3 report development-period (2019–2020) benchmark diagnostics. The 2019–2020 partition participated in cohort construction and informed model selection, so these values are development diagnostics rather than an independent test; they are reported because they are the basis on which the model suite and the comparison set were fixed, and they anticipate the held-out evaluation. Unless stated otherwise, all development values are five-seed ensemble means on the 249,072 common keys, with station-median RMSE in °C. Section 4.4 reports the held-out 2021–2023 comparative evaluation; all of its values are unweighted station medians over the reportable stations on the common held-out key registry (`outputs/final/`), the same station-first estimator used in Sections 4.1–4.3. Pooled RMSE is reported only as a sensitivity in the Supporting Information and is never labelled a station median.

4.1 The reference model, not the architecture, sets the reported gain

How much of a reported gain survives a strong reference? Most of it does not. On identical keys, the same fixed model reports a seven-day median station skill of +0.251 against persistence and +0.038 against damped persistence — a factor of 6.6 between two numbers that differ only in the denominator of equation (3).

Lead	Persistence	Damped persistence	LightGBM	LSTM	ThermoRoute
1 d	0.803	0.774	0.578	0.662	0.631
3 d	1.576	1.406	1.280	1.323	1.291
7 d	2.217	1.738	1.649	1.679	1.657

Development-period (2019–2020) station-median RMSE, °C. Read down the seven-day row: station-median RMSE falls from 2.217 °C under persistence to 1.738 °C under damped persistence to 1.657 °C under ThermoRoute. The median over stations of the per-station memory gain (persistence minus damped persistence) is 0.479 °C and the median learned gain (damped persistence minus ThermoRoute) is 0.081 °C; these are medians of per-station differences and, like all such quantities, do not add exactly to the difference of the station-median RMSEs (Section 3.6). The same arithmetic in skill units gives +0.203, +0.187, and +0.251 against persistence and +0.168, +0.076, and +0.038 against damped persistence at 1, 3, and 7 days, all dimensionless.

The gap widens with lead, which is the diagnostic signature of damping rather than of learned river behavior: the longer the lead, the more of the error a seasonal relaxation removes on its own, and the less remains for a learned model to remove. A study reporting only the persistence column would describe this model as gaining a quarter of the seven-day error budget. A study reporting both columns describes the same fitted weights as gaining four percent.

Reported against the strong reference, the development-period paired effects of equation (2) are −0.130, −0.104, and −0.062 °C at 1, 3, and 7 days, with ThermoRoute favored at 0.86, 0.90, and 0.93 of the 120 stations, and whole-HUC2 cluster-bootstrap intervals of [−0.177, −0.081], [−0.126, −0.083], and [−0.071, −0.052] °C. Those intervals carry the 15-cluster structure discussed in Section 3.6 and are approximate. The held-out 2021–2023 evaluation (Section 4.4) reports the same comparison on the test window. If the reference set determines the size of the reported gain, the next question is whether it also determines which model wins.

Figure 3. Decomposition of reported skill on the held-out 2021–2023 window (116 reportable stations, common forecast-key registry). (a) Station-median RMSE at 1-, 3-, and 7-day leads for persistence, damped persistence, LightGBM, LSTM, and ThermoRoute; lower is better. (b) Seven-day error-budget decomposition: per-station memory gain (persistence minus damped persistence) and learned gain (damped persistence minus ThermoRoute), summarized as the equal-station mean waterfall (mean memory gain 0.456 °C plus mean learned gain 0.070 °C equals the mean total gain 0.527 °C exactly) with the median and IQR of

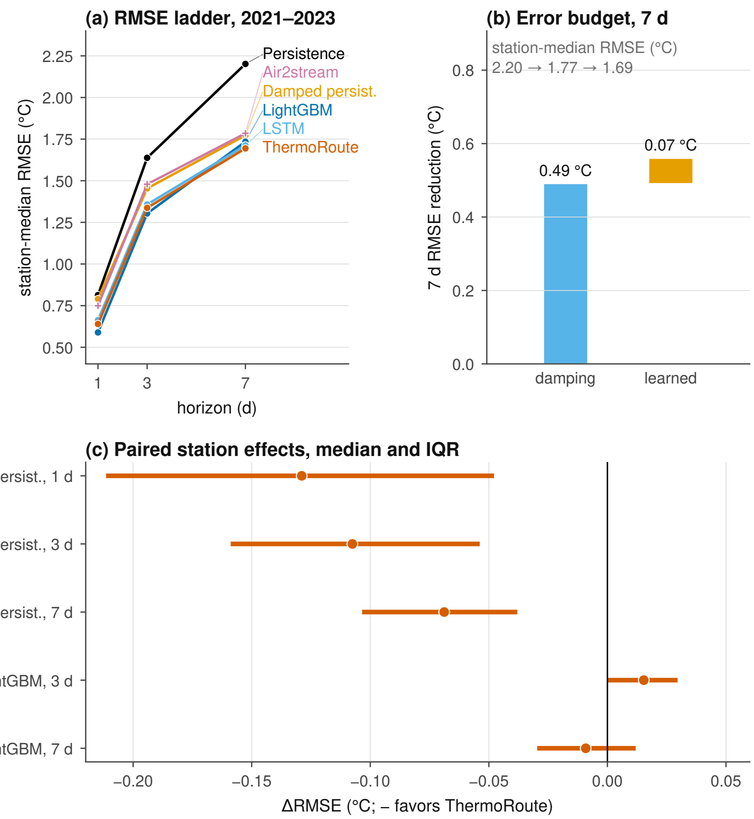

Figure 2. Decomposition of reported skill on the held-out window.

the per-station gains shown alongside (station-median memory gain 0.491 °C and learned gain 0.069 °C at seven days). (c) Per-station memory fraction (memory gain divided by total gain, computed only where the total gain is positive; median 0.875 at seven days). (d) Paired station-level ΔRMSE (ThermoRoute minus reference), unweighted medians with IQR across 116 stations; negative values favor ThermoRoute; clustered intervals, win rates, and Holm-adjusted p-values are in Table 4.6.

4.2 The most accurate model on this panel is a gradient-boosted tree

It does. On the development window the tree ensemble has the lowest station-median RMSE at all three leads: 0.578, 1.280, and 1.649 °C against ThermoRoute's 0.631, 1.291, and 1.657 °C. The paired station-level differences (`ThermoRoute` − `LightGBM`, equation 9) are +0.046 °C at 1 day, +0.008 °C at 3 days, and −0.005 °C at 7 days, with ThermoRoute win rates of 0.00, 0.40, and 0.60 across the 120 stations. The one-day win rate is 0.00: not one of the 120 stations favors the constrained architecture at the shortest lead.

We report this directly. On this panel, on identical keys, a well-tuned gradient-boosted tree with site identity is more accurate than the constrained deep architecture, decisively at 1 day and negligibly at 3 and 7, where the cluster-bootstrap intervals for the paired difference are [+0.001, +0.020] and [−0.010, +0.010] °C. The one-day paired difference is outside the frozen five-test family; as an exploratory check it is +0.035 °C (ThermoRoute minus LightGBM, station median, HUC2-cluster bootstrap CI [+0.031, +0.045], sign-flip p = 1.0 under the one-sided family convention), with the tree favored at 98% of the 116 reportable stations — the abstract's one-day claim is therefore paired, cluster-aware, and consistent with the development-window win rate of 0.00. The global LSTM is intermediate at 0.662, 1.323, and 1.679 °C. The architecture's remaining practical argument is not accuracy but the algebraic bound of equation (7), which held on 100.00% of audited rows, with a maximum absolute correction of 1.0000 °C against the configured 1 °C limit and the derived station-by-lead RMSE inequality holding in all 360 cells — a property of the construction rather than a fitted outcome.

4.3 An information-matched plain convolutional network reproduces the architecture

Very little of the architecture's structure is doing work. Against the information-matched plain causal temporal convolutional network (same inputs, same anchor, same optimiser budget, 38,346 against 38,505 parameters) the full architecture's paired median station effect, computed within seed on identical keys, ranges across the five seeds from −0.0003 to −0.0105 °C at 1 day and −0.0035 to −0.0226 °C at 7 days — every seed favors the full model, none by more than 0.023 °C. The one-factor deletions agree: removing the temporal encoder is the largest single effect (1-day station-median RMSE 0.631 to 0.679 °C), while removing the router, mixture, dynamic prior, fixed-relaxation constraint, or residual bound moves the 1-day figure by at most 0.004 °C (held-out ablation values in SI09). Against the plain multilayer perceptron the same effects are −0.036 to −0.041 °C at 1 day, so sequence structure matters and the components layered on top of it do not. These are five-seed deletion and intervention sensitivities on paired keys; they do not prove component necessity.

4.4 Held-out 2021–2023 evaluation

The held-out 2021–2023 evaluation applies the model suite and comparison set of Section 3, fixed on data through 2020, to the test window 2021-01-01 through 2023-12-31. Every number in this section is an unweighted station median over the reportable stations on the common held-out key registry, derived from `outputs/final/` by `scripts/final/build_results_authority.py` (pooled RMSE appears only as a Supporting Information sensitivity, Section SI07). On this window the deep model's station-median skill is +0.250 against persistence and +0.038 against damped persistence, and the tree with site identity leads at one and three days (station-median RMSE 0.589 and 1.304 °C; Table 4.7a) and is matched at seven days (1.694 versus 1.735 °C; paired station-median ΔRMSE −0.009 °C, Table 4.6 row 5). The information-matched plain neural controls are scored on this window with the same frozen weights and keys as every other model; they carry no calibration (`NO_FROZEN_CALIBRATION`) and enter only the point comparisons.

Table 4.6 — paired comparisons on the held-out window. Station-level ΔRMSE (°C, negative favors ThermoRoute), computed as the unweighted median over reportable stations of RMSE_ThermoRoute − RMSE_reference, for the five formal tests of Section 3.6 (the frozen five-test family). CI low and CI high are the 2.5% and 97.5% percentiles of a cluster bootstrap that resamples whole HUC2 regions; the win rate is the fraction of reportable stations where the candidate has lower RMSE. p (sign flip) is the whole-HUC2-cluster sign-flip p-value and Holm p is its Holm adjustment over the five formal tests (Section 3.6). Rows are generated by `scripts/final/generate_manuscript_tables.py` from `outputs/final/`.

#	Comparison	Lead	ΔRMSE (°C)	CI low	CI high	Win rate	Stations	p (sign flip)	Holm p
1	ThermoRoute vs. Damped-Persistence	1	-0.129	-0.199	-0.076	0.90	116	3.1e-05	1.5e-04
2	ThermoRoute vs. Damped-Persistence	3	-0.108	-0.143	-0.073	0.91	116	3.1e-05	1.5e-04
3	ThermoRoute vs. Damped-Persistence	7	-0.069	-0.086	-0.057	0.95	116	6.1e-05	1.8e-04
4	ThermoRoute vs. LightGBM	3	0.015	0.010	0.025	0.24	116	1.0e+00	1.0e+00
5	ThermoRoute vs. LightGBM	7	-0.009	-0.017	0.002	0.59	116	7.4e-02	1.5e-01

Tables 4.7–4.8 — held-out 2021–2023 metrics. Station-median metrics per model and lead over the reportable stations ($n = 116$ at every lead): RMSE, MAE, and bias in °C; skill against persistence and damped persistence (equation 10, dimensionless, positive favors the candidate). The information-matched plain controls are included on the same station set and denominators; the ablations are deletion and intervention sensitivities and do not prove component necessity. Skill is the unweighted median over reportable stations of the per-station ratio $1 -$

RMSE_model/RMSE_baseline; a ratio of station-median RMSEs is a different quantity and is not used. Rows are generated by `scripts/final/generate_manuscript_tables.py` from `outputs/final/`.

Model	RMSE 1	RMSE 3	RMSE 7	MAE 1	MAE 3	MAE 7	bias 1	bias 3	bias 7
Persistence	0.813	1.638	2.202	0.597	1.235	1.686	-0.000	-0.001	-0.004
DampedPersistence	0.789	1.454	1.773	0.591	1.100	1.340	-0.029	-0.076	-0.143
Climatology	1.899	1.902	1.903	1.485	1.486	1.485	-0.379	-0.369	-0.359
LightGBM	0.589	1.304	1.735	0.431	0.995	1.315	-0.025	-0.098	-0.196
LSTM	0.663	1.358	1.712	0.503	1.044	1.292	-0.001	-0.051	-0.137
PlainMLP-7var	0.690	1.374	1.720	0.520	1.054	1.311	-0.004	-0.059	-0.128
PlainCausalTCN-7var	0.646	1.330	1.710	0.477	1.021	1.299	0.000	-0.031	-0.122
Air2stream	0.719	1.459	1.825	0.559	1.115	1.366	-0.020	-0.110	-0.189
ThermoRoute	0.640	1.337	1.694	0.463	1.013	1.269	0.011	-0.029	-0.130

The development-period benchmark diagnostics of Sections 4.1–4.3 anticipate the held-out evaluation. On the held-out window the one-factor ablations neither help nor hurt systematically at any lead: fixing the dynamic prior (fixed κ) or removing the bounded-residual constraint (unbounded) yields station-median skill against damped persistence of +0.180/+0.176 at one day against +0.173 for the full model, with small, non-monotonic differences that never exceed +0.009 skill units across the leads and tables (SI09). The dynamic prior and the bounded-residual constraint therefore add no material point-accuracy gain on the independent window; this is precisely the class of negative result that a weak evaluation design would hide.

Two of those findings have held-out counterparts in the 2021–2023 cells above: the reference model rather than the architecture sets the reported gain — seven-day skill against persistence remains high while skill against damped persistence collapses — and a gradient-boosted tree with site identity has the lowest station-median RMSE at the one- and three-day leads on identical held-out keys. The fitted air2stream-style hybrid (Toffolon and Piccolroaz, 2015) — an unofficial empirical thermal-recurrence variant of the published formulation, not the official code and not a validated reproduction — reaches a station-median RMSE of 0.719 °C at one day, the best non-learned result on the common 116-station set, and falls behind damped persistence at 3 and 7 days (1.459 and 1.825 °C); because the implementation is not the published model, no process-side claim is attached to it. The information-matched plain convolutional network and plain multilayer perceptron are re-scored on this held-out window (Tables 4.7a and SI07): the plain TCN's station-median RMSE is 0.646, 1.330, and 1.710 °C at 1, 3, and 7 days against ThermoRoute's 0.640, 1.337, and 1.694 °C — the full architecture retains a small, consistent edge (median paired ΔRMSE with the plain TCN as candidate and ThermoRoute as reference, +0.010, +0.011, and +0.015 °C; under equation (2) positive values favor the reference, and the plain-TCN win fractions are 0.30, 0.30, and 0.16) that is an order of magnitude smaller than the reference-model effect, and the plain MLP is worse still (0.690, 1.374, and 1.720 °C), so sequence structure matters while the router, mixture, and bounding machinery add little beyond the causal encoder and the anchor.

4.5 Matched spatial-transfer experiment on 2021–2023

The development-period whole-region finding (SI19) is re-tested on the independent window with a matched 2×2 factorial design that separates spatial geometry from local adaptation (protocol v1; details in SI11). Station-agnostic gradient-boosted trees (site identity removed, frozen per-lead hyperparameters, deterministic fits) are fitted on the in-fold stations' 2006–2017 rows under two geometries — four deterministic leave-HUC2-region folds and four balanced random folds repeated over five split seeds — and two adaptation policies: *local*, in which each held station's own 2006–2015 history contributes to its climatology, damped anchor and imputation medians (gauged transfer with local thermal history), and *pooled*, in which those statistics are fitted over the in-fold training stations only, so a held region's long histories never enter preprocessing. Each held-out station is scored on its own 2021–2023 common forecast keys; both a raw-target and a damped-anchor-residual variant are fitted.

Under local adaptation the median nearest-training-gauge distance is 60 km for the random arm and 263 km for the whole-region arm. The region-minus-random paired station penalty (region cell minus the mean of the five random-split cells, per station) is +0.006 °C at one day (IQR −0.000 to +0.014), +0.009 °C at three days (IQR +0.001 to +0.026), and +0.007 °C at seven days (IQR +0.001 to +0.030) for the raw-target tree, with the random arm winning at 72–78% of stations and the penalty positive in all five split seeds at every lead (+0.004 to +0.010 °C per seed). The residual-target tree gives the same picture (+0.006, +0.007, and +0.004 °C). The penalty is of the same order as the random-split spread (median per-site standard deviation across the five seeds, 0.004–0.007 °C) and is an order of magnitude smaller than the reference-model effect of Section 4.1; within this design there is no strong gradient of the penalty against distance or hydroclimatic novelty (correlations −0.18 to +0.17 across leads and models). Whole-region holdout therefore imposes a small, consistently signed transfer penalty that is reproducible across models, seeds, and leads, but the spatial partition is a secondary effect relative to the reference model (decision rule R1 in the protocol). The pooled-adaptation cells, which exclude a held region's long histories from preprocessing, do not share the local cells' geometry ordering: pooled-cell errors are much larger at the same geometry (region-pooled 7-day station-median RMSE 2.051 versus 1.725 °C local; random-pooled 1.824 versus 1.700 °C local for the raw-target tree), and the 7-day region-minus-random paired penalty changes sign within the pooled arm (−0.004 °C, 47% of stations favoring the region cell) against +0.007 °C (76%) in the local arm (SI11). On the station sets matched across all four cells, the adaptation main effect (pooled minus local, seven days) is +0.20 °C with 96% of stations consistently signed — an order of magnitude above the geometry effect — while the geometry main effect within the local arm is +0.005 °C; the spatial partition is therefore secondary specifically given local thermal history. These pooled-arm estimates are a lower bound: the pooled

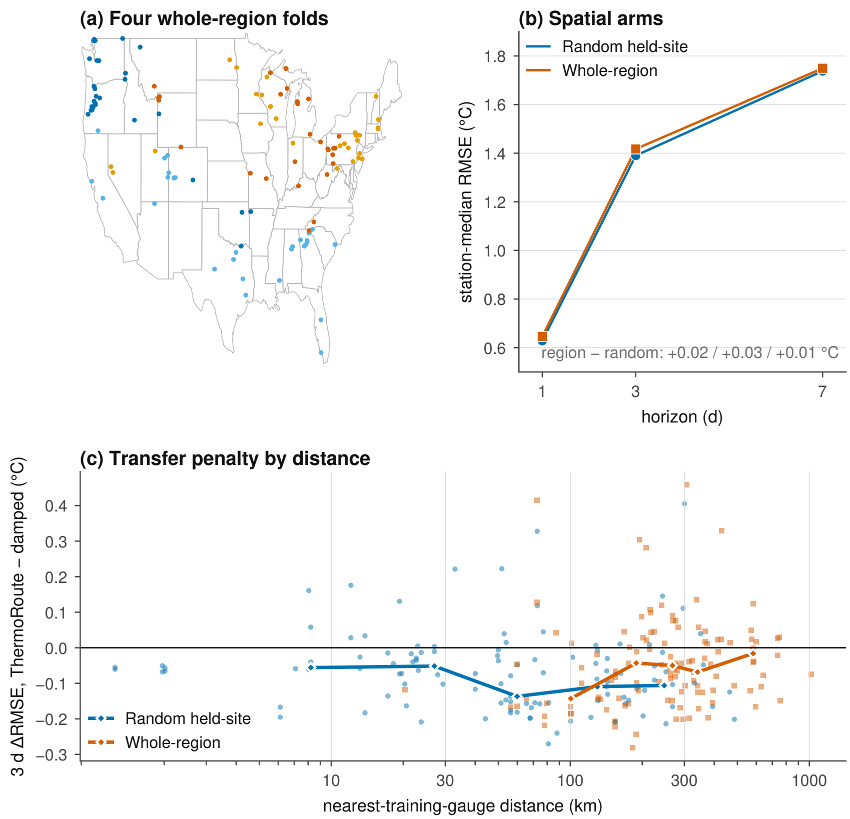

Figure 3. Spatial transfer under matched random-site and whole-region holdouts.

preprocessing of this run reused fold 0's statistics across folds (a recorded defect, DLOG-012), so the true cost of losing local history is larger; protocol v2 reruns the design with per-fold preprocessing and adds levels with no local record at all.

Figure 5. Spatial transfer under matched random-site and whole-region holdouts on the independent 2021–2023 window. (a) The four whole-HUC2-region folds on a CONUS outline. (b) Station-median RMSE for the four factorial cells (geometry × adaptation) at 1-, 3-, and 7-day leads (station-agnostic LightGBM, identical preprocessing and keys within each cell). (c) Paired region-minus- random three-day error as a function of distance to the nearest training gauge; each point is one station–seed cell, positive values favor the random arm, and bin medians are marked with diamonds. (d) The same penalty against a standardized hydroclimatic-novelty distance to the nearest training station. The penalty is small (about 0.01 °C), consistently signed across five split seeds and two model targets, and does not show a strong novelty gradient (cell-level station-set caveats for the pooled arm are recorded in SI11).

4.6 Hydrologic conditions governing incremental skill

The remaining learned gain is concentrated in specific hydrologic states, which locates the mechanism that survives the strong reference. Station thermal memory is measured as the anomaly half-life of the official train-fitted damped-persistence anchor, $t_{1/2} = \ln(0.5)/\ln(\phi_i)$ — an anomaly half-life, not an e-folding time — with a station median of 6.9 days (interquartile range 5.0–11.9). The station-level memory/learned decomposition of the error budget is consistent with it: at seven days the median memory gain is 0.49 °C and the median learned gain 0.07 °C (Figure 3b; per-station fractions, computed where the total gain is positive, have a median memory fraction of 0.875). The station correlation between log half-life and learned gain is −0.40 at one day and −0.14 at three days — longer-memory stations leave less for the learned model to add at short leads — and reverses sign at seven days (+0.21), where the remaining learned gain is small everywhere.

Issue-time-identifiable states. All thresholds are station-specific (or station-month-specific) empirical quantiles fitted on the 2006–2015 training period only; state metrics are computed station-first with a 30-key minimum per station-state cell, then summarized by the median across stations (Table 4.12; details in SI11). At seven days the learned model's median paired ΔRMSE over damped persistence is −0.069 °C on all keys (Table 4.6, row 3). It is largest — in favor of the learned model — when the current thermal anomaly is low (−0.23 °C, 105 stations) and during rapid recent cooling (−0.17 °C) and recent warming (−0.09 °C); the gain is present under both low and high station-relative flow (−0.07 and −0.06 °C) and across all seasons (−0.05 to −0.10 °C). No issue-time-identifiable state reverses the ordering: the learned correction pays for itself under every state examined here, but the magnitude of its advantage is state-dependent rather than uniform.

Retrospective outcome-conditioned diagnostics. Stratifying by the target outcome — an explicitly retrospective diagnostic, not an issue-time state — shows where the anchor is weakest: on days that subsequently warm rapidly (actual change at or above the station's training q90, positive direction only) the median paired ΔRMSE reaches −0.30 °C at seven days, and on days that subsequently cool rapidly (training q10 or below) it is −0.18 °C. On the coldest target decile the anchor alone is marginally better (+0.03 °C), the only stratum in which the learned correction does not pay for itself.

Table 4.12 — Hydrologic states (7-day keys). Median across stations of the per-station paired ΔRMSE (ThermoRoute − damped persistence, °C; negative favors ThermoRoute), station count, and median keys per station-state cell. `issue_*` states are issue-time-identifiable with station-specific training-period (2006–2015) quantiles (station-month quantiles for flow); `actual_*` states are retrospective outcome-conditioned diagnostics with station-specific signed thresholds. Rows are generated by `scripts/final/generate_manuscript_tables.py`.

State (7-day keys)	ΔRMSE, ThermoRoute − damped (°C)	stations	median keys
All keys	-0.069	116	1,060
issue low anomaly	-0.229	105	71
issue high anomaly	-0.080	114	139
issue rapid recent warming	-0.088	116	107
issue rapid recent cooling	-0.170	114	99
issue strong warming pressure	-0.072	115	111
issue strong cooling pressure	-0.058	116	90
issue low flow	-0.072	92	143
issue high flow	-0.060	104	89
issue rapid flow rise	-0.054	116	103
issue rapid flow recession	-0.072	116	99
actual rapid warming	-0.300	116	109
actual rapid cooling	-0.184	116	102
actual warmest decile	-0.066	116	141
actual coldest decile	0.029	109	77
actual high flow	-0.078	107	87
actual low flow	-0.078	92	147

Reported average skill therefore understates a state-dependent benefit that a strong reference obscures but does not remove; the benefit is largest on days that subsequently warm or cool rapidly, while the issue-time signature of those days — recent trend, anomaly, disequilibrium — is present but weaker.

Figure 6. Hydrologic conditions governing incremental skill (held-out 2021–2023). (a) Distribution of the station anomaly half-life from the official train-fitted damped-persistence anchor (median 6.9 d; an anomaly half-life, not an e-

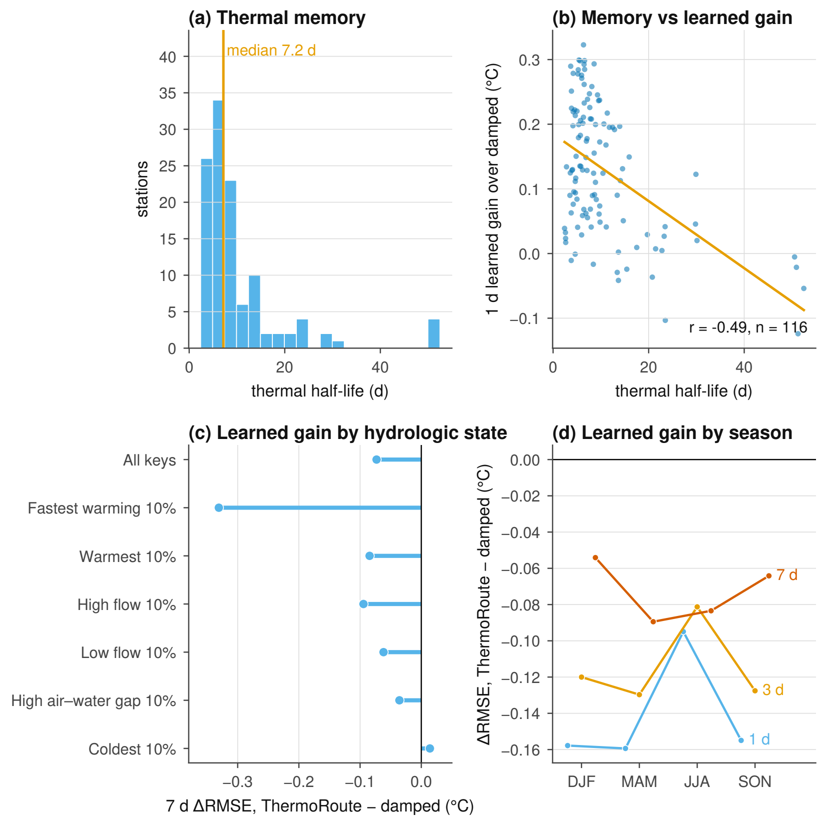

Figure 4. Hydrologic conditions governing incremental skill.

folding time). (b) One-day learned gain over damped persistence versus log half-life, with the least-squares trend (r = −0.40); longer-memory stations leave less to learn at short leads. (c) Seven-day median station paired ΔRMSE (ThermoRoute minus damped persistence) within issue-time-identifiable states; negative values favor ThermoRoute. (d) Retrospective outcome-conditioned strata (actual rapid warming and cooling computed separately with station-specific signed thresholds); negative values favor ThermoRoute. The learned model's incremental value concentrates in rapid warming and cooling states and under low thermal anomaly; it is absent on the coldest target decile.

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5 Discussion

5.1 Why these numbers are smaller than published gains

Our reported gains are far smaller than those commonly published for daily river temperature, and the difference is a difference of design rather than of model quality. Three specific choices account for it, and each can be read off Section 4 directly.

The first is the reference. Where a study quotes skill against persistence or climatology, the number it reports is comparable to our +0.203 to +0.251 column, not to our +0.038 to +0.168 column. On a variable with this much day-to-day memory the two differ by up to a factor of 6.6 for the same fitted weights, and the discrepancy grows with lead, so studies at longer leads are the ones most affected. We would expect much of the between-study dispersion noted by Corona and Hogue (2025) to be absorbed by re-scoring published models against a fitted damped anchor rather than against persistence, and we suggest damped persistence as the minimum reference for this variable.

The second is the spatial partition. A model evaluated on randomly held-out gauges in a dense network is being asked to interpolate between instrumented neighbours; the same model evaluated on whole held-out regions, roughly 260 km from the nearest training gauge, loses about 0.01 °C of station-median RMSE relative to random-site transfer on the independent window — a small but reproducibly signed penalty that is an order of magnitude smaller than the reference-model effect. Where published transfer results use random site splits, the quantity they measure is closer to interpolation than to transfer, but on this cohort the magnitude of the difference is modest.

The third is the key set and the preprocessing boundary. Scoring each model on its own complete-case set and fitting scaling, imputation, climatology, event thresholds, or interval offsets on windows that include the evaluated interval both move error downward without leaving a visible trace. We cannot quantify how much of the published literature is affected, and we do not assert that it is; what we can say is that removing both channels here left the constrained deep model behind a gradient-boosted tree at the 1- and 3-day leads and matched at 7 days.

We are more cautious about the mechanistic reading of Section 4.3. That an information-matched plain causal convolutional network reproduces the full architecture to within 0.023 °C, and that deleting the router, the mixture, the dynamic prior, or the residual bound moves 1-day error by at most 0.004 °C, suggests that the accuracy available to a point-scale statistical predictor on this panel is largely exhausted by a causal sequence encoder over issue-time history plus a seasonal anchor. It does not identify why. Capacity matching is not search-budget matching, deletion is not attribution, and the controls are diagnostic throughout.

5.2 What a benchmark of this geometry can carry

The evaluation design of this study is stronger than its inferential reach, and the two should not be confused. Clustered inference over a gauge panel rests on resampling or sign-flipping whole spatial units, and the accuracy of those procedures depends on having enough units, of comparable size, that the asymptotics they invoke are not a fiction. Fifteen HUC2 groups with an inverse-Herfindahl effective count of 9.54 and a largest group holding 21.7% of stations is not enough, and the sampling design was never exchangeable across regions. We therefore present the clustered intervals and p-values as approximate descriptive sensitivities and do not attach decision weight to them.

The usual practice in applied water-temperature machine learning is to report a station-level or row-level confidence interval, or a paired test across sites, without addressing whether the number of independent spatial units can carry the procedure. Under that practice a cohort like ours would produce intervals and p-values that look authoritative, and nothing in the paper would tell a reader that the effective number of clusters is under ten. The difference between that paper and this one is not the data and not the model; it is whether the limit is disclosed.

The design lesson is transferable. If a study intends region-clustered inference over U.S. gauges, the cluster structure must be part of the *sampling design*, not a post-hoc grouping of whatever cohort availability produced, and the partition and its power should be fixed before the evaluation is read. Reaching a genuinely large number of independent spatial units requires either a finer partition with a defensible independence argument or a deliberately stratified draw across many more regions; neither is achievable by relabelling an existing cohort.

5.3 What transfers

This study is narrow by construction. It is a common-key, clustered comparative benchmark of point predictors at gauged sites, and it is not an attempt to replace process-based thermal models, river-network graph models, or differentiable hybrid formulations (Jia et al., 2021; Rahmani et al., 2023; Zwart et al., 2023), nor architectures that encode geographic context across regions and scales (Luo et al., 2025), each of which represents structure — connectivity, upstream forcing, energy balance, spatial context — that a point-scale statistical predictor does not. Every arm here, including the whole-region and site-identifier-disjoint arms, consumes the target site's observed water temperature through the issue date (Section 3.4), so the paper is silent on prediction at locations with no thermal record.

What is portable is not the architecture but three controls that do not depend on it: scoring every model on an identical key registry so that no model benefits from selective prediction; fitting every preprocessing and calibration statistic strictly backwards in time; and partitioning space by whole hydrologic regions rather than by site. None requires unusual computational resources, and the reference choice alone moved a reported number in this study by far more than the differences between the tested architectures. On this evidence, benchmark design is a larger source of variation in reported river-temperature skill than model design is, with the caveat that the key-selection and preprocessing channels are closed by construction in this design rather than quantified as separate effects (Section 4.5, SI11).

5.4 Limitations

All caveats are consolidated here; earlier sections refer to them by number.

Scope. The design is a descriptive benchmark on a fixed cohort: comparisons are station-first and approximate (15 HUC2 clusters; effective cluster count 9.54; largest share 21.7%; non-probability sampling), and no interval, p-value, or ranking claim is decision evidence. Prediction at locations with no water-temperature record is not quantified in the current evidence package: the earlier L2/L3 and forcing-ladder learned results were withdrawn after the raw-observedness lineage defect was identified, and the corrected score-producing authorities remain pending. The manuscript therefore makes no operational-replay or ungauged-location performance claim from those withdrawn runs. The fitted relaxation rate is a statistical quantity only, not a heat-transfer coefficient, and no physical interpretation of its value is offered. Throughout the paper ”causal” describes time ordering only, never causal inference.

Cohort, measurement, and design. The cohort trades a real, modest discharge channel (removal costs +0.042 °C at 1 day; SI19) against wide spatial coverage; 950 of 1,465 candidates were excluded for missing joint flow. Point-scale meteorology is used at station coordinates rather than basin integrals. The air2stream-style hybrid is an unofficial empirical thermal-recurrence variant of the published formulation (Toffolon and Piccolroaz, 2015) — not the official code, not a validated reproduction — so no process-side claim is attached to its scores; running the official code is planned (protocol v3, Phase 5b). Training and inference wall-clock costs were not recorded. The development-period spatial analysis and robustness probes of the earlier draft are archived in SI19.

Threshold and archive scope. The ±1 °C algebraic bound, event-threshold quantiles, and reportability thresholds (100 paired keys) are development-selected; the q90 event threshold is a statistical tail diagnostic with no biological, ecological, or regulatory interpretation. Public data licenses and rights are discussed in SI16.

6 Conclusions

We evaluated daily river water-temperature prediction at 120 stable U.S. gauges across 15 hydrologic regions, scoring every model on one common set of station/date/horizon keys at 1-, 3-, and 7-day leads, with every preprocessing and calibration statistic fitted strictly backwards in time and with a held-out 2021–2023 test window for the comparative evaluation.

Three findings survive on that independent window. First, the reference model governs the reported gain: the deep predictor's seven-day station-median skill is +0.250 against persistence but +0.038 against damped persistence, with a median station-level memory gain of 0.49 °C and a median learned gain of 0.07 °C at seven days (the per-station memory fraction has a median of 0.875). Second, model ranking does not favor architectural constraint: a gradient-boosted tree with site identity has the lowest station-median RMSE at 1- and 3-day leads (0.589 and 1.304 °C), and the deep model's 7-day point estimate (1.694 versus 1.735 °C) is not distinguishable from the tree at the cluster level (Table 4.6); the one-factor ablations (router, mixture, dynamic prior, residual bound) move one-day error by at most 0.004 °C on the development window with no material held-out gain. Third, the spatial partition affects reported transfer only when the held station's own thermal history is unavailable: development-period three-day skill against persistence falls from +0.187 under a random held-site split to +0.116 under whole-region holdout, while on the independent window the matched factorial shows a small, consistently signed region penalty of about 0.01 °C — reproducible across models and split seeds but an order of magnitude below the reference-model effect (Section 4.5). On the same independent window, withholding the station's own thermal history from preprocessing (pooled adaptation) costs roughly 0.2 °C at seven days — about 40 times the geometry effect at that lead — and the pooled cells do not share the local cells' geometry ordering (the 7-day region-minus-random penalty changes sign), so the spatial partition is secondary specifically *given* local thermal history (protocol v2 extends this into a four-level information ladder that includes prediction with no local water-temperature record at all).

The common mechanism is that each design choice absorbs an alternative explanation into the reported number: seasonal damping into a weak reference, selective prediction into a model-specific key set, and spatial interpolation into a random site split. For water-resource practice, reported skill is what a manager compares when choosing a tool, so a published skill score is not interpretable without its reference model and its spatial partition, and the incremental value of architectural complexity over a strong statistical anchor and a well-tuned tree should be re-measured under these controls before it is relied upon.

The comparisons are descriptive for this fixed, availability-selected cohort: the clustered intervals and p-values are approximate sensitivities (at most 15 HUC2 groups), not decision evidence, and no result generalizes to a national or U.S.-river population.

Open Research Statement

Data availability. The derived daily panel (`panel_usgs_120v2.parquet`), the stable station registry (`station_registry_v1.csv`), the hydrologic-unit metadata snapshot, the candidate-rejection ledger, and the panel manifest that binds them are deposited as one versioned dataset at `[DATA DOI TO BE MINTED]` under `[DATA LICENCE TO BE ASSIGNED]`. The test-window acquisition record — exact request and response bytes, series identifiers, approval qualifiers, final URLs, retrieval timestamps, and content hashes — is added to that deposit as a new version.

Neither the DOI nor the data licence can be assigned before the byte-level rights review described in Section 6.4 completes. The derived panel encodes values from three providers whose terms differ, and a derived product does not inherit the most permissive of its upstream terms. Until every object proposed for the deposit carries a recorded, evidence-backed redistribution decision, the deposit is described here as planned and specified, not as existing.

Primary observational sources. All observations are obtained from public providers and none is the property of the authors. Daily-value water temperature and discharge come from the U.S. Geological Survey National Water Information System (U.S. Geological Survey, 1994; parameter codes 00010, 00060, and 00065 with statistic code 00003). Air temperature, precipitation, the relative-humidity proxy, and daylight-period mean incoming shortwave radiation come from Daymet V4 R1 at ORNL DAAC (Thornton et al., 2022). Wind speed comes from gridMET (Abatzoglou, 2013). Each retains its provider's own terms, citation requirement, and access route independently of this manuscript.

Material that cannot currently be redistributed. For any provider object whose redistribution terms are unresolved at deposit time, the deposit carries, in place of the bytes: the product identifier and version, the exact request specification, the retrieval code, and a SHA-256 manifest of the bytes as retrieved, so a third party can re-acquire identical inputs and verify them without relying on redistribution. The classes currently defaulting to exclusion, and the two excluded outright — the third-party AGU LaTeX class, and three legacy CSV files retained in the repository archive from an earlier three-station case study whose source and collection terms are unrecorded — are enumerated in SI16; the legacy material is not evidence for any statement in this manuscript. Original provider responses for the 2006–2020 development panel were not retained, so development reproduction begins from the committed derived artifact; that limitation does not apply to the test-window acquisition, for which exact bytes are archived.

Software availability. The analysis software, the model-suite registry, the per-stage manifests, the environment probe, the verification entrypoints, the result renderer that generates Section 4.4 from the test-window metrics, and the fully transitive Python 3.12 dependency lock with package hashes are archived at `[SOFTWARE DOI TO BE MINTED]`, corresponding to release `[RELEASE TAG TO BE ASSIGNED]` of the source repository at `[REPOSITORY URL TO BE CONFIRMED]`. The project's own source code is released under the MIT licence (SPDX identifier `MIT`). That licence covers the source code only: it does not license the observational data, the archived provider responses, the third-party typesetting class, or any redistributed dependency binary, each of which retains its own terms and is enumerated with those terms in an archive notice file.

Analyses were run on Python 3.12 with NumPy, pandas, pyarrow, SciPy (Virtanen et al., 2020), scikit-learn (Pedregosa et al., 2011), LightGBM (Ke et al., 2017), and PyTorch (Paszke et al., 2019). USGS NWIS daily values are retrieved by direct requests to the USGS waterservices daily-values REST endpoint, with every request and response byte-pair cached and content-addressed by the in-repository `SnapshotStore` provenance store.

Reproduction. Sections 4.1–4.3 are reproducible from the archived panel, the archived model bundles, and the pinned environment. Section 4.4 is reproducible from the archived test-window acquisition record and the same bundles. Bit-level equality is expected for the tree models and agreement to a documented tolerance for the neural members; the tolerances, the verification commands, and the expected digests are listed in the Supporting Information. Reproduction has not yet been carried out by an operator independent of the authors on an independent host, and this statement is not a claim that it has.

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[COMPUTATIONAL RESOURCES TO BE COMPLETED — the facility or facilities on which the model suite was fitted.]

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Competing interests. [COMPETING INTERESTS TO BE COMPLETED — a declaration is required from every author, including the explicit statement that none exists if that is the case.]

Author contributions. [CREDIT ROLES TO BE COMPLETED — assign each author to the applicable CRediT roles: conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing — original draft, writing — review and editing, visualization, supervision, project administration, funding acquisition. The signed intake form is `docs/FAIR_SUBMISSION_READINESS_AND_TEMPLATES.md` §2.]

Supporting Information

Supporting Information accompanies this manuscript and contains: the cohort and registry description with the candidate-rejection ledger (SI01); the issue-time information boundary and the product-compatibility bridge (SI02); model equations and unit conventions (SI03); the analysis protocol, redesign chronology and decision log (SI04); the comparison set and its fields (SI05–SI06); all-model scores on the exact common keys, including the fitted air2stream-style hybrid reference with its parameter ranges and official-variant caveat (SI07); probability and reliability schemas and the full measured degenerate-interval table (SI08); architecture and information-matched controls with seed and budget provenance and their independent-window re-test (SI09); the temporal coverage audit (SI10); the matched spatial factorial, local-adaptation policy, repeated-split distribution and the cluster geometry at HUC2, HUC4, HUC6, and HUC8 (SI11); outcome quality control and qualifier evidence (SI12); the history-dependent external arm (SI13); missingness and failure cases with the key-history completeness strata (SI14); reproduction hashes, commands, and environment parity fields, including the results-authority commands and manifest (SI15); and the rights and data dictionary (SI16).

Supporting figures accompany these sections: Figures S1–S3 describe the cohort geometry, the temporal roles and issue-time information boundary, and the model and calibration dataflow; Figures S4–S8 and S10 expand Section 4.4; and Figure S9 is a development-period conformal-calibration sensitivity, labeled as such in the figure itself, which must not be compared numerically with any held-out-window figure.

Each figure carries evidence from exactly one period. Figures 1 and S1–S3 are structural material; Figures 3, 5, and 6 and Figures S4–S8 and S10 are held-out-window; Figure S9 is development-period (Figure 4 includes the held-out re-test of the information-matched controls in Section 4.6) and say so inside the figure. No value from one period is compared numerically with a value from another.

References

Abatzoglou, J. T. (2013). Development of gridded surface meteorological data for ecological applications and modelling. *International Journal of Climatology*, *33*, 121–131. (gridMET / METDATA) doi: 10.1002/joc.3413

Bai, S., Kolter, J. Z., & Koltun, V. (2018). *An empirical evaluation of generic convolutional and recurrent networks for sequence modeling.* (arXiv:1803.01271)

Barclay, J. R., Topp, S. N., Koenig, L. E., Sleckman, M. J., & Appling, A. P. (2023). Train, inform, borrow, or combine? approaches to process-guided deep learning for groundwater-influenced stream temperature prediction. *Water Resources Research*, *59*, e2023WR035327. doi: 10.1029/2023WR035327

Beven, K., & Binley, A. (1992). The future of distributed models: model calibration and uncertainty prediction. *Hydrological Processes*, *6*(3), 279–298. doi: 10.1002/hyp.3360060305

Caissie, D. (2006). The thermal regime of rivers: a review. *Freshwater Biology*, *51*, 1389–1406. doi: 10.1111/j.1365-2427.2006.01597.x

Corona, C. R., & Hogue, T. S. (2025). Machine learning in stream and river water temperature modeling: a review and metrics for evaluation. *Hydrology and Earth System Sciences*, *29*, 2521–2549. doi: 10.5194/hess-29-2521-2025

Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, *13*, 253–263. doi: 10.1080/07350015.1995.10524599

Feigl, M., Lebiedzinski, K., Herrnegger, M., & Schulz, K. (2021). Machine-learning methods for stream water temperature prediction. *Hydrology and Earth System Sciences*, *25*, 2951–2977. doi: 10.5194/hess-25-2951-2021

Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, *102*, 359–378. doi: 10.1198/016214506000001437

Gupta, H. V., Kling, H., Yilmaz, K. K., & Martinez, G. F. (2009). Decomposition of the mean squared error and NSE performance criteria: implications for improving hydrological modelling. *Journal of Hydrology*, *377*, 80–91. doi: 10.1016/j.jhydrol.2009.08.003

Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, *13*, 281–291. doi: 10.1016/S0169-2070(96)00719-4

Hodson, T. O., & Hariharan, J. A. (2023). *dataretrieval (Python): a Python package for discovering and retrieving water data available from federal hydrologic web services.* U.S. Geological Survey software release. doi: 10.5066/P94I5TX3

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, *6*(2), 65–70. Retrieved from `https://www.jstor.org/stable/4615733`

Jia, X., Zwart, J., Sadler, J., Appling, A., Oliver, S., Markstrom, S., … Kumar, V. (2021). Physics-guided recurrent graph model for predicting flow and temperature in river networks. In *Proceedings of the 2021 siam international conference on data mining (sdm)* (pp. 612–620). SIAM. (Preprint: arXiv:2009.12575) doi: 10.1137/1.9781611976700.69

Kavetski, D., Kuczera, G., & Franks, S. W. (2006a). Bayesian analysis of input uncertainty in hydrological modeling: 1. theory. *Water Resources Research*, *42*, W03407. doi: 10.1029/2005WR004368

Kavetski, D., Kuczera, G., & Franks, S. W. (2006b). Bayesian analysis of input uncertainty in hydrological modeling: 2. application. *Water Resources Research*, *42*, W03408. doi: 10.1029/2005WR004376

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., … Liu, T.-Y. (2017). LightGBM: a highly efficient gradient boosting decision tree. In *Advances in neural information processing systems 30 (neurips 2017)* (pp. 3146–3154).

Retrieved from https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html

Kingma, D. P., & Ba, J. (2015). Adam: a method for stochastic optimization. In *International conference on learning representations (iclr).*

Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, *17*, 1217–1241. doi: 10.1214/aos/1176347265

Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. In *International conference on learning representations (iclr).*

Luo, S., Yu, R., Chen, S., Fan, Y., Xie, Y., Li, Y., & Jia, X. (2025). Geo-aware models for stream temperature prediction across different spatial regions and scales. In *Proceedings of the 33rd acm international conference on advances in geographic information systems (sigspatial '25)* (pp. 124–136). ACM. (Preprint: arXiv:2510.09500) doi: 10.1145/3748636.3762716

Martins, A. F. T., & Astudillo, R. F. (2016). From softmax to sparsemax: a sparse model of attention and multi-label classification. In *Proceedings of the 33rd international conference on machine learning (icml)* (pp. 1614–1623). Retrieved from https://proceedings.mlr.press/v48/martins16.html

Mohseni, O., Stefan, H. G., & Erickson, T. R. (1998). A nonlinear regression model for weekly stream temperatures. *Water Resources Research*, *34*, 2685–2692. doi: 10.1029/98WR01877

Nash, J. E., & Sutcliffe, J. V. (1970). River flow forecasting through conceptual models part I — a discussion of principles. *Journal of Hydrology*, *10*, 282–290. doi: 10.1016/0022-1694(70)90255-6

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., ... Chintala, S. (2019). PyTorch: an imperative style, high-performance deep learning library. In *Advances in neural information processing systems 32 (neurips 2019)* (pp. 8024–8035).

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... Duchesnay, É. (2011). Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, *12*, 2825–2830.

Piccolroaz, S., Calamita, E., Majone, B., Gallice, A., Siviglia, A., & Toffolon, M. (2016). Prediction of river water temperature: a comparison between a new family of hybrid models and statistical approaches. *Hydrological Processes*, *30*, 3901–3917. doi: 10.1002/hyp.10913

Rahmani, F., Appling, A., Feng, D., Lawson, K., & Shen, C. (2023). Identifying structural priors in a hybrid differentiable model for stream water temperature modeling. *Water Resources Research*, *59*, e2023WR034420. doi: 10.1029/2023WR034420

Rahmani, F., Lawson, K., Ouyang, W., Appling, A., Oliver, S., & Shen, C. (2021). Exploring the exceptional performance of a deep learning stream temperature model and the value of streamflow data. *Environmental Research Letters*, *16*, 024025. doi: 10.1088/1748-9326/abd501

Rahmani, F., Shen, C., Oliver, S., Lawson, K., & Appling, A. (2021). Deep learning approaches for improving prediction of daily stream temperature in data-scarce, unmonitored, and dammed basins. *Hydrological Processes*, *35*, e14400. doi: 10.1002/hyp.14400

Richardson, D. S. (2000). Skill and relative economic value of the ECMWF ensemble prediction system. *Quarterly Journal of the Royal Meteorological Society*, *126*, 649–667. doi: 10.1002/qj.49712656313

Romano, Y., Patterson, E., & Candès, E. J. (2019). Conformalized quantile regression. In *Advances in neural information processing systems 32 (neurips 2019)* (pp. 3538–3548). Retrieved from `https://papers.nips.cc/paper/2019/hash/5103c3584b063c431bd1268e9b5e76fb-Abstract.html`

Sadler, J. M., Appling, A. P., Read, J. S., Oliver, S. K., Jia, X., Zwart, J. A., & Kumar, V. (2022). Multi-task deep learning of daily streamflow and water temperature. *Water Resources Research*, *58*, e2021WR030138. doi: 10.1029/2021WR030138

Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., & Dean, J. (2017). Outrageously large neural networks: the sparsely-gated mixture-of-experts layer. In *International conference on learning representations (iclr).* Retrieved from `https://arxiv.org/abs/1701.06538`

Sun, A. Y., Jiang, P., Mudunuru, M. K., & Chen, X. (2021). Explore spatio-temporal learning of large sample hydrology using graph neural networks. *Water Resources Research*, *57*, e2021WR030394. doi: 10.1029/2021WR030394

Thornton, M. M., Shrestha, R., Wei, Y., Thornton, P. E., & Kao, S.-C. (2022).

Daymet: daily surface weather data on a 1-km grid for North America, version 4 R1. ORNL DAAC, Oak Ridge, Tennessee, USA. doi: 10.3334/ORNLDAAC/2129

Toffolon, M., & Piccolroaz, S. (2015). A hybrid model for river water temperature as a function of air temperature and discharge. *Environmental Research Letters*, *10*, 114011. doi: 10.1088/1748-9326/10/11/114011

Topp, S. N., Barclay, J., Diaz, J., Sun, A. Y., Jia, X., Lu, D., … Appling, A. P. (2023). Stream temperature prediction in a shifting environment: explaining the influence of deep learning architecture. *Water Resources Research*, *59*, e2022WR033880. doi: 10.1029/2022WR033880

United States Geological Survey. (1994). *USGS water data for the nation.* National Water Information System (NWIS) daily values accessed via the dataretrieval Python client; parameter codes 00010, 00060, 00065 with statistic code 00003. (Accessed 2026; `https://waterdata.usgs.gov/nwis`) doi: 10.5066/F7P55KJN

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., … SciPy 1.0 Contributors (2020). SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, *17*, 261–272. doi: 10.1038/s41592-019-0686-2

Vovk, V., Gammerman, A., & Shafer, G. (2005). *Algorithmic learning in a random world.* New York: Springer. doi: 10.1007/b106715

Weierbach, H., Lima, A. R., Willard, J. D., Hendrix, V. C., Christianson, D. S., Lubich, M., & Varadharajan, C. (2022). Stream temperature predictions for river basin management in the pacific northwest and mid-atlantic regions using machine learning. *Water*, *14*, 1032. doi: 10.3390/w14071032

Wilks, D. S. (2011). *Statistical methods in the atmospheric sciences* (3rd ed., Vol. 100). Oxford: Academic Press. doi: 10.1016/c2010-0-65519-2

Zwart, J. A., Diaz, J., Hamshaw, S., Oliver, S., Ross, J. C., Sleckman, M., … White, E. (2023). Evaluating deep learning architecture and data assimilation for improving water temperature forecasts at unmonitored locations. *Frontiers in Water*, *5*, 1184992. doi: 10.3389/frwa.2023.1184992